

Math261 Pre-Quiz5 Review

Part A: Conceptual / Visualization

A1. Sketch the cross section (in the xz -plane, i.e. $y = 0$) of the solid bounded by the cylinder $x^2 + y^2 = 4$, above by the plane $z = 6 - x$, and below by the plane $z = 0$. Label where the boundary curves cross the z -axis and the vertical lines $x = -2$ and $x = 2$.

A2. Sketch the cross section (in the xz -plane, i.e. $y = 0$) of the solid bounded above by $z = \sqrt{x^2 + y^2}$ and below by $z = x^2 + y^2$. Label where the two curves intersect, and shade the region between them.

A3. For the solid $\rho \leq 2$, $\pi/6 \leq \varphi \leq \pi/3$, $0 \leq \theta \leq 2\pi$, describe in words what the solid looks like (e.g., is it a full cone, a wedge, a shell, a band around the sphere?).

A4. True or false, and justify briefly: “In spherical coordinates, φ always ranges over 0 to 2π , just like θ does.” If false, state the correct range and explain the geometric reason for the difference.

Part B. Error Analysis (Spot the Mixed-Up Coordinates)

Each item below sets up an integral that mixes coordinate systems incorrectly, or has a mismatched volume element. Identify the error(s) and correct them.

B1.

$$\iiint_E f \, dV = \int_0^{2\pi} \int_0^2 \int_0^{4-r^2} f(r, \theta, z) \cdot \rho^2 \sin \varphi \, dz \, dr \, d\theta$$

B2.

$$\iiint_E f \, dV = \int_0^{2\pi} \int_0^\pi \int_0^a f(\rho, \theta, \varphi) \, d\rho \, d\varphi \, d\theta$$

B3. A student is asked to find the volume of the solid ball $x^2 + y^2 + z^2 \leq 4$ and writes:

$$V = \int_0^{2\pi} \int_0^2 \int_0^2 r \cdot \rho^2 \sin \varphi \, dr \, d\rho \, d\theta$$

Identify every problem with this setup (there is more than one).

B4. A student converts the cone $z = \sqrt{x^2 + y^2}$ into spherical coordinates and gets $\rho = \pi/4$. Explain the conceptual mistake (not just “wrong formula”) that led to this error.

Part C. Set Up Only (Do Not Evaluate)

C1. Set up the integral, in cylindrical coordinates, for the volume of the solid bounded above by $z = 8 - x^2 - y^2$ and below by $z = x^2 + y^2$.

C2. Set up the integral, in spherical coordinates, for $\iiint_E (x^2 + y^2 + z^2) \, dV$, where E is the region between the spheres $\rho = 1$ and $\rho = 3$.

C3. Set up the integral, in cylindrical coordinates, for the volume of the solid that lies inside the cylinder $x^2 + y^2 = 1$, above the plane $z = 0$, and below the cone $z = 2 - \sqrt{x^2 + y^2}$.

C4. Set up the integral, using whichever coordinate system is most natural, for the mass of the solid hemisphere $x^2 + y^2 + z^2 \leq 9$, $z \geq 0$, with density $\delta(x, y, z) = \sqrt{x^2 + y^2 + z^2}$.

Part D. Short Computation

D1. Evaluate $\int_0^{2\pi} \int_0^1 \int_0^r z \cdot r \, dz \, dr \, d\theta$.

D2. Evaluate $\int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta$.

D3. Find the volume of the solid cylinder $x^2 + y^2 \leq 1$, $0 \leq z \leq 3$, by setting up and evaluating a triple integral in cylindrical coordinates. (This should confirm the elementary formula $V = \pi r^2 h$.)

D4. Evaluate $\iiint_E z^2 \, dV$ where E is the solid ball $\rho \leq 1$, using spherical coordinates.

Answer Key (Brief)

A1: The cross section is bounded below by the horizontal line $z = 0$ and above by the tilted line $z = 6 - x$, over $-2 \leq x \leq 2$ (from the cylinder). At $x = -2$, the top boundary is at $z = 8$; at $x = 2$, it's at $z = 4$; the two boundary lines don't meet within this strip, so the cross section is a trapezoid-shaped region, not a triangle.

A2: In the xz -plane ($y = 0$), the cone becomes $z = |x|$ and the paraboloid becomes $z = x^2$. These meet where $|x| = x^2$, i.e., at $x = -1, 0, 1$. Between $x = -1$ and $x = 1$, the cone lies above the paraboloid, so the cross section is the lens-shaped region between the two curves, pinched to a point at the origin.

A3: A spherical “band” or zone: a shell-like sector of the ball of radius 2, restricted to angles between 30° and 60° from the positive z -axis, swept through a full revolution in θ — visually like a cone-shaped band cut out of a solid sphere.

A4: False. φ ranges over 0 to π only, because φ measures the angle down from the positive z -axis, and any point in 3D space can be reached with φ between 0 (straight up) and π (straight down); going past π would just retrace points already covered by adjusting θ instead.

B1: The volume element mixes cylindrical variables (r, θ, z) with the spherical factor $\rho^2 \sin \varphi$. It should simply be $r \, dz \, dr \, d\theta$.

B2: Missing the $\rho^2 \sin \varphi$ factor entirely — the spherical volume element is $\rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta$, not $d\rho \, d\varphi \, d\theta$.

B3: Multiple errors: (i) it mixes r (cylindrical) and ρ (spherical) in the same integral; (ii) it's missing $\sin \varphi$; (iii) the bounds 0 to 2 for both r and ρ double-count/misassign the radius — a ball of radius 2 should just be set up in spherical alone, as $0 \leq \rho \leq 2$, $0 \leq \varphi \leq \pi$, $0 \leq \theta \leq 2\pi$, with volume element $\rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta$.

B4: The student converted the equation as if setting ρ equal to the angle rather than solving for φ ; the cone $z = \sqrt{x^2 + y^2}$ becomes $z = r$, i.e., $\rho \cos \varphi = \rho \sin \varphi$, giving $\tan \varphi = 1$, so $\varphi = \pi/4$ — a constant *angle*, not a value of ρ . The mistake is conflating “the equation simplifies to a constant” with “that constant is ρ ” instead of recognizing which spherical variable actually becomes constant.

$$\mathbf{D1:} \quad 2\pi \int_0^1 \int_0^r z r \, dz \, dr = 2\pi \int_0^1 r \cdot \frac{r^2}{2} \, dr = 2\pi \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{\pi}{4}.$$

$$\mathbf{D2:} \quad \frac{\pi}{2} \cdot [-\cos \varphi]_0^{\pi/2} \cdot \left[\frac{\rho^3}{3} \right]_0^1 = \frac{\pi}{2} \cdot 1 \cdot \frac{1}{3} = \frac{\pi}{6}.$$

$$\mathbf{D3:} \quad V = \int_0^{2\pi} \int_0^1 \int_0^3 r \, dz \, dr \, d\theta = 2\pi \cdot 3 \cdot \frac{1}{2} = 3\pi = \pi(1)^2(3). \quad \checkmark$$

$$\mathbf{D4:} \quad \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^2 \cos^2 \varphi \cdot \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta = 2\pi \cdot \frac{1}{5} \cdot \int_0^\pi \cos^2 \varphi \sin \varphi \, d\varphi = 2\pi \cdot \frac{1}{5} \cdot \frac{2}{3} = \frac{4\pi}{15}.$$